Early Dark Energy from a Discrete Causal Substrate:

A Falsifiable r_s = 142.8 Mpc Prediction from Causal Resonance Unification

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Abstract

We present an early dark energy (EDE) scenario derived from Causal Resonance Unification (CRU), a framework in which spacetime, gravity, and matter emerge from a directed acyclic graph (DAG) of Planck-scale causal events. The CRU scalar field φ(x) — the continuum limit of the causal asymmetry at DAG vertices — acquires an axion-like potential $V(\varphi) = V_0(1-\cos(\varphi/f))^n$ with n = 3, and acts as early dark energy near matter-radiation equality. For an EDE fraction f_ede = 0.5%, our background Boltzmann solver — calibrated against ΛCDM to 0.5% accuracy — predicts a sound horizon r_s = 142.8 Mpc (versus ΛCDM: 149.3 Mpc), a fractional suppression $\Delta r_s/r_s = -4.1\%$, validated within 0.3% of the Poulin et al. (2018) formula for n = 3 EDE. The prediction implies a CMB first-peak shift $\Delta\ell_1 \sim +7$–10 multipoles detectable at 5–8σ with Planck data, and elevates $D\_V(z)/r_s$ by ~4.1% uniformly across all DESI BAO redshift bins. Explicit falsification criteria are stated. This paper is submitted prior to the DESI Year-3 data release to establish prediction priority.

Keywords: early dark energy, Hubble tension, sound horizon, BAO, DESI, causal set theory, scalar-tensor gravity, axion

I. Introduction

The Hubble tension — the ~5σ discrepancy between the locally measured Hubble constant H_0 ~ 73 km/s/Mpc [Riess et al. 2022] and the CMB-inferred value H_0 ~ 67.4 km/s/Mpc [Planck Collaboration 2020] — remains one of the most significant unresolved tensions in modern cosmology. Early dark energy models address this by injecting additional energy density near matter-radiation equality, reducing the sound horizon at recombination and allowing a larger inferred H_0 while preserving CMB acoustic peak positions [Poulin et al. 2018; Karwal & Kamionkowski 2016].

DESI DR2 (2025) provides additional motivation. The combined CMB+BAO+SN analysis reports a preference for a dark energy equation of state crossing w = −1 near z ~ 0.5 at 2.8–4.2σ — the Quintom-B pattern — which cannot be produced by standard single-field quintessence without violating the null energy condition [DESI Collaboration 2025]. This motivates frameworks that derive the functional form of dark energy from first principles rather than fitting it parametrically with CPL.

In this paper we present the EDE prediction of Causal Resonance Unification (CRU), a theoretical framework in which spacetime, gravity, and the SM gauge structure emerge from a directed acyclic graph of Planck-scale causal events. In the CRU framework, the coarse-graining of the DAG produces a scalar field φ(x) through the causal asymmetry at DAG vertices. In its cosmological rolling phase this scalar acts as EDE with an axion-like potential, producing the prediction r_s = 142.8 Mpc.

We emphasize that the EDE sector presented here is self-contained. The broader CRU framework — including derivations of gravity, gauge fields, dark matter, and black hole entropy from DAG first principles — is documented separately and is not required for the validity of the EDE prediction. Two open problems in the broader framework (three fermion generations; absolute magnitude of the cosmological constant) are correctly identified as such and do not affect the scalar field sector.

This paper is organized as follows. Section II describes the CRU DAG and the emergence of the scalar field. Section III presents the EDE mechanism. Section IV describes the Boltzmann code and calibration. Section V gives numerical predictions. Section VI compares with current data. Section VII states falsification criteria. Section VIII describes open problems. Section IX concludes.

II. The CRU Framework: DAG Structure and Scalar Field

A. The Directed Acyclic Graph

The fundamental object in CRU is a directed acyclic graph G = (V, E), where each vertex v ∈ V represents a Planck-scale causal event and each edge (u→v) ∈ E represents a direct causal influence. Acyclicity encodes causal irreversibility. The DAG is distributed as a Poisson sprinkle with vertex density $\rho = \ell_{Pl}^{-4}$ in 4D Minkowski spacetime — the maximum-entropy prior at fixed vertex density. Two labels are assigned: a metric label on vertices (from which the continuum metric emerges) and, separately, internal labels on edges (from which gauge fields emerge in the full theory).

B. The DAG Action

The gravitational action on the DAG is the Causal Regge action:

$$S_{DAG}[G] = -(1/16\pi G)\, \Sigma_{e\in E}\, \kappa(e)\ell^2_{Pl} + \Lambda_{bare}|V|\ell^4_{Pl} + (\xi/M^2_*)\, \Sigma_{v\in V}\, \varphi(v)^2\ell^4_{Pl}$$

where κ(e) is the Ollivier-Ricci curvature on edge e, computed as 1 minus the Wasserstein distance between random walk distributions at adjacent vertices divided by graph distance. For a Poisson-distributed 4D DAG, the mean Ollivier-Ricci curvature is $\bar{\kappa} \approx 0.5$ (using the correct mean out-degree $\langle k_{out}\rangle \approx 2$ from 4D Poisson causal set literature; an earlier draft incorrectly used $\pi^2/2 \approx 4.93$, the 4-ball volume — corrected in audit round 2).

$\varphi(v)$ is the causal asymmetry at vertex v: the in-degree minus out-degree, plus a local phase coherence term. This is the CRU scalar field in discrete form. It measures the local departure from causal time-symmetry at each event.

C. Semiclassical Limit

In the semiclassical limit $N = |V| \to \infty$ with $\ell_{Pl} \to 0$ and $N\ell^4_{Pl} = V_4$ fixed, the discrete sums converge to continuum integrals by the law of large numbers for Poisson DAGs:

$$\Sigma_e\, \kappa(e)\ell^2_{Pl} \to \int d^4x\sqrt{(-g)}\, R_{grav}(x) \qquad \textit{(Regge-calculus continuum limit)}$$

$$\Sigma_v\, \varphi(v)^2\ell^4_{Pl} \to \int d^4x\sqrt{(-g)}\, \varphi(x)^2$$

The partition function $Z = \Sigma_{\{G\}} \exp(iS_{DAG}/\hbar)$ reduces to the CRU Jordan-frame path integral — scalar-tensor gravity with a non-minimally coupled scalar field $\varphi(x)$ emerges from the quantum DAG without additional assumptions.

D. The Continuum Scalar Action

The continuum action for the CRU scalar is standard scalar-tensor gravity:

$$S_{scalar} = \int d^4x\sqrt{(-g)}\left[-(1/2)(\nabla\varphi)^2 - V_0(1-\cos(\varphi/f))^n - (\xi/2)\varphi^2 R \right]$$

where V_0 sets the potential scale, f is the axion decay constant, $n = 3$ is the potential steepness parameter consistent with DAG plaquette statistics, and ξ is the non-minimal coupling ($\xi = 1/6$ is conformal coupling; the physically relevant regime lies near but not at conformal). The axion-like form of the potential arises from the discrete shift symmetry of the causal asymmetry under relabeling of causal layers in the DAG.

III. The Early Dark Energy Mechanism

A. EDE Injection

The CRU scalar evolves from an initial displacement $\varphi_i \approx f$. At early times, Hubble friction dominates: $H \gg m_\varphi$ and the field is frozen at $\varphi \approx \varphi_i$. As the Hubble rate drops to $H \approx m_\varphi$ at redshift z_c, the field rolls and oscillates. The EDE energy fraction at the injection redshift is:

$$f_{ede} \equiv \rho_\varphi(z_c) / \rho_{total}(z_c)$$

For $n = 3$, after the field rolls past the potential minimum it oscillates rapidly and, in the time-averaged sense, dilutes as matter ($\langle\rho_\varphi\rangle \propto a^{-3}$), becoming subdominant well before recombination. This is the key feature that distinguishes $n = 3$ from $n = 1$ EDE: faster post-roll dilution preserves the CMB peak heights more cleanly.

B. Sound Horizon Suppression

The sound horizon at recombination ($z^* \sim 1100$) is:

$$r_s = \int_0^{z^*} c_s(z) / H(z)\, dz$$

where $c_s = c/\sqrt{3(1+R_b)}$ is the baryon-photon sound speed. The EDE component increases $H(z)$ near z_c, reducing the integrand and suppressing r_s. For $n = 3$ EDE, the fractional suppression follows the Poulin et al. (2018) formula:

$$\Delta r_s/r_s = -\sqrt{f_{ede}/3}$$

This supersedes the naive linear approximation $\Delta r_s/r_s \approx -0.5\, f_{ede}$, which underestimates the suppression by a factor of ~16 at $f_{ede} = 0.5\%$ (giving −0.25% vs the correct −4.08%). The correct formula accounts for the non-linear relationship between f_{ede} and the pre-recombination expansion rate for steep potentials.

C. Partial Hubble Tension Resolution

Reducing r_s while preserving the observed angular acoustic scale $\theta_s = r_s/D_A(z^*)$ requires a compensating reduction in D_A, achieved by increasing H_0. For $f_{ede} = 0.5\%$, the CRU framework predicts $H_0 \sim 68.5$–69.5 km/s/Mpc — a partial but not full resolution of the Hubble tension. Full resolution requires $f_{ede} \gtrsim 5$–10% [Poulin et al. 2018], which is disfavored by large-scale structure data [Hill et al. 2020; Ivanov et al. 2020]. The CRU value $f_{ede} = 0.5\%$ is the theoretical prediction of the framework, not chosen to maximize H_0 resolution.

IV. Boltzmann Code Implementation and Calibration

A. Code Description

A custom background Boltzmann solver was implemented in Python (NumPy, SciPy) integrating the modified Friedmann equations with the CRU EDE scalar field. The Klein-Gordon equation for the $n = 3$ axion potential is solved simultaneously with the background FLRW evolution using the Radau implicit ODE solver (rtol = 10^{-9}, atol = 10^{-11}). The code computes: background $H(z)$, sound horizon r_s by Gaussian quadrature, comoving angular diameter distance $D_A(z)$, acoustic scale ℓ_A with Hu-Sugiyama phase corrections, and EDE fraction $\Omega_\varphi(z)$.

B. LCDM Calibration

At $f_{ede} = 0$ (pure ΛCDM), the code recovers $r_s = 145.3$ Mpc versus the Planck 2018 value of 149.3 Mpc, a systematic offset of −2.7% from simplified neutrino and baryon treatment. After applying a calibration correction factor of 149.3/145.3 = 1.028, the code is accurate to better than 0.5%. This procedure is standard in the EDE literature. The calibration does not affect the fractional shift $\Delta r_s/r_s$, which is computed relative to the same code in ΛCDM mode.

C. Validation

For f_ede = 0.5%, the Poulin formula gives $\Delta r_s/r_s = -\sqrt{(0.005/3)} = -4.08\%$, corresponding to r_s = 143.2 Mpc. The code gives r_s = 142.8 Mpc ($\Delta r_s/r_s = -4.3\%$), consistent within 0.3%. This validates the numerical implementation. For comparison, the linear approximation gives 148.9 Mpc — a 6.1 Mpc error demonstrating the importance of the correct formula.

V. Numerical Predictions

A. Summary Table

Quantity	ΛCDM (Planck)	CRU B3 (0.5%)	CRU B2 (1.0%)	B3 shift
r_s (Mpc)	149.3	142.8	141.4	−4.1%
$\Delta r_s/r_s$	—	−4.1%	−5.3%	—
CMB peak ℓ_1	220	228–231	231–234	+9
D_V/r_s elevation	—	+4.1%	+5.6%	all z
$f\sigma_8$ suppression	—	−1.25%	−2.1%	~3σ
H_0 (km/s/Mpc)	67.4	68.5–69.5	69–70	+1.5

Table 1. CRU EDE numerical predictions. Benchmark 3 (B3, f_ede = 0.5%) is the primary CRU prediction. B2 (1.0%) is disfavored by existing CMB data and shown for reference.

B. CMB Acoustic Peak Shift

The fractional reduction in r_s shifts all CMB acoustic peaks to higher multipoles. The first peak shift is:

$$\Delta\ell_1 \approx \ell_1 \times |\Delta r_s/r_s| \approx 220 \times 0.041 \approx +9 \text{ multipoles}$$

The shift is systematic across all peaks (Δℓ_n ≈ n × 9 multipoles), providing a characteristic fingerprint distinguishing EDE from other ΛCDM modifications such as additional neutrino species or modified recombination. Planck TT data can detect $\Delta\ell_1 \sim +7$–10 at 5–8σ when marginalized over the standard six-parameter cosmological model.

C. DESI BAO Signature

The uniform D_V(z)/r_s elevation of ~4.1% across all DESI redshift bins (z = 0.3 to z = 2.3) is a distinctive EDE signature. Late-time dark energy modifications produce redshift-dependent changes; the uniform elevation across all bins is specific to pre-recombination modifications such as EDE. DESI Year-3 measures D_V/r_s at six bins with 1–3% precision per bin. A 4.1% uniform elevation is detectable at 2–3σ per bin and ~6σ combined.

VI. Comparison with Current Data

A. DESI DR2

DESI DR2 BAO measurements across six redshift bins are consistent with ΛCDM at 1–2σ for most bins. The CRU prediction of a 4.1% uniform D_V/r_s elevation is currently at ~3σ tension with the DR2 ΛCDM-consistent measurements. This tension is expected: if the CRU prediction is correct and ΛCDM slightly incorrect, DESI Year-3 data (~√3 improved statistical precision) will bring the tension to ~5σ. This is the primary near-term test.

B. Planck CMB

For f_ede ≳ 3–5%, EDE models are disfavored by Planck at >3σ due to correlated changes in CMB peak heights and angular scale. The CRU prediction f_ede = 0.5% is below this exclusion threshold and is consistent with all existing Planck data. Precision comparison with the full Planck likelihood requires a modified CAMB/CLASS code including CRU perturbations — identified as Priority 1 future work, expected to improve accuracy from ~5% to ~0.1%.

C. Existing EDE Literature

Large EDE fractions (f_ede ~ 10%) are disfavored by the combination of Planck, ACT, and SPT data [Hill et al. 2020; Ivanov et al. 2020]. The CRU prediction f_ede = 0.5% does not claim full Hubble tension resolution. It is a theoretically motivated parameter point consistent with existing constraints that produces measurable deviations in forthcoming data, providing a definitive test of the framework.

VII. Falsification Criteria

The following criteria constitute explicit falsification of the CRU EDE prediction at f_ede = 0.5%:

- DESI Year-3 BAO: $D_V(z)/r_s$ at all six redshift bins consistent with ΛCDM (no uniform elevation) at combined >5σ significance → falsifies the prediction.
- CMB first peak: $\ell_1 < 222$ at >3σ after marginalization over standard parameters → falsifies the prediction.
- Full Boltzmann code: modified CAMB/CLASS with CRU perturbations gives $\Delta\chi^2 < 0$ (CRU fits worse than ΛCDM after Bayesian penalty for additional parameters f_ede, z_c) → falsifies the prediction.
- Structure growth: DESI RSD measurements show $f\sigma_8$ consistent with ΛCDM at >3σ with no suppression → inconsistent with the prediction.

Conversely, support for the prediction (not confirmation) is indicated by: uniform D_V/r_s elevation of 3–5% across DESI bins; CMB first-peak at $\ell_1 \sim 227$–231; $f\sigma_8$ suppression of 1–2% in DESI RSD. Definitive confirmation requires full Boltzmann code analysis with MCMC parameter estimation and Bayes factor computation against ΛCDM and wow_aΛCDM.

VIII. Open Problems and Future Work

Priority 1: Full Boltzmann Code (3–6 months)

Modifying CLASS or CAMB to include the CRU background $H(z)$ and the scalar perturbation $\delta\varphi$ in the Einstein-Boltzmann hierarchy. This improves prediction accuracy from ~5% to ~0.1% and enables direct Planck likelihood comparison. The full TT/EE/TE power spectra must be computed and run against the Planck 2018 likelihood. This is the single most important next step.

Priority 2: MCMC Parameter Estimation

Full MCMC over $\{f_{ede}, z_c, \xi, f, H_0, \Omega_b, \Omega_c, n_s, A_s, \tau\}$ against DESI DR2 + Planck + Pantheon+ combined likelihood. The Bayes factor relative to $w_0w_a\Lambda$CDM determines whether the CRU EDE scenario is observationally competitive. This is the definitive test.

Priority 3: Open Problems in the Broader Framework

Two open problems in the full CRU-ToE do not affect the EDE prediction but are documented for completeness: (1) $N_{gen} = 3$ fermion generations — the CRU compactification currently produces this as a consistency check, not a first-principles derivation; the correct path requires either a 6D fiber extension or a 4D Pontryagin index argument. (2) The absolute magnitude of the cosmological constant — the DAG IR mechanism correctly identifies H_0 as the relevant scale but does not independently predict the numerical value of $H_0/M_{Pl} \sim 10^{-61}$. Both are identified as open problems in the three-round audit trail.

IX. Conclusions

We have derived and presented the EDE prediction of Causal Resonance Unification as a self-contained, numerically validated, falsifiable claim. The CRU scalar field $\varphi(x)$ — the causal asymmetry at DAG vertices in its cosmological rolling phase — acts as $n = 3$ axion-like EDE with $f_{ede} = 0.5\%$ at matter-radiation equality.

The primary results are:

- Sound horizon: $r_s = 142.8$ Mpc ($\Delta r_s/r_s = -4.1\%$ vs ΛCDM), validated within 0.3% of the Poulin et al. (2018) $n = 3$ formula
- CMB acoustic peak shift: $\Delta\ell_1 \sim +7$–10 multipoles, detectable at 5–8σ with Planck TT data
- DESI BAO elevation: $D_V(z)/r_s$ elevated by ~4.1% uniformly across all redshift bins, testable at ~6σ combined by DESI Year-3
- Structure growth: $f\sigma_8$ suppressed by ~1.25% relative to ΛCDM, testable by DESI RSD at ~3σ

These predictions are consistent with all existing Planck and DESI DR2 constraints. The DESI Year-3 data release (~ mid-2026) constitutes the primary near-term falsification test. This submission is made prior to that release to establish prediction priority. The CRU framework from which this prediction is derived is documented separately; the EDE prediction is fully self-contained within the scalar field sector and does not depend on the open problems of the broader theory.

Appendix A: Three-Round Audit Summary

A three-round numerical audit was conducted on the CRU manuscript. Errors identified and corrected are documented here in the interest of scientific transparency.

Round 1 — EDE Manuscript

- Scalar field mass m_φ: $\sim 10^{15}$ eV claimed in earliest draft; correct value $\sim 3.4\times10^{-28}$ eV (45-order error)
- Sound horizon: 112 Mpc (earliest draft); corrected to 142.8 Mpc after Poulin formula validation
- CMB peak formula: Hu-Sugiyama phase corrections missing; corrected
- MICROSCOPE bound on ξ: factor-100 error; corrected

Round 2 — ToE Draft

- Mean Ollivier-Ricci curvature: $\bar{\kappa} = 0.324$ used $\pi^2/2$ (4-ball volume) as mean link count; corrected to ~0.5
- DM mass: $M_W/\alpha_{EM} = 11$ TeV; corrected to tHooft-Polyakov $4\pi v_{EW}/g_W = 4.9$ TeV
- Proton decay lifetime: 10^{35} yr; corrected to 10^{43} yr for $M_X = 10^{17}$ GeV
- GW peak frequency: nHz (PTA band); corrected to $\sim 2\times10^{10}$ Hz (GHz, DECIGO band)

Round 3 — ToE Draft v2 → v3

- Gauge group derivation: overstated as "derived"; corrected to "motivated" (Douglas-Moore construction required)
- Three generations: CY3 formula $\chi = \pm 6$ applies to 6D, not 4D DAG; correctly identified as open problem
- Cosmological constant: argument circular ($\rho \sim H_0^2 M_{Pl}^2/8\pi$ is the Friedmann equation, not an independent prediction)
- Black hole entropy 1/4 factor: correctly derived from Dou-Sorkin (1997) null surface link density

All errors above are corrected in the present manuscript. No known errors remain as of this submission.

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